Hypointense Findings on Hepatobiliary Phase MR Images

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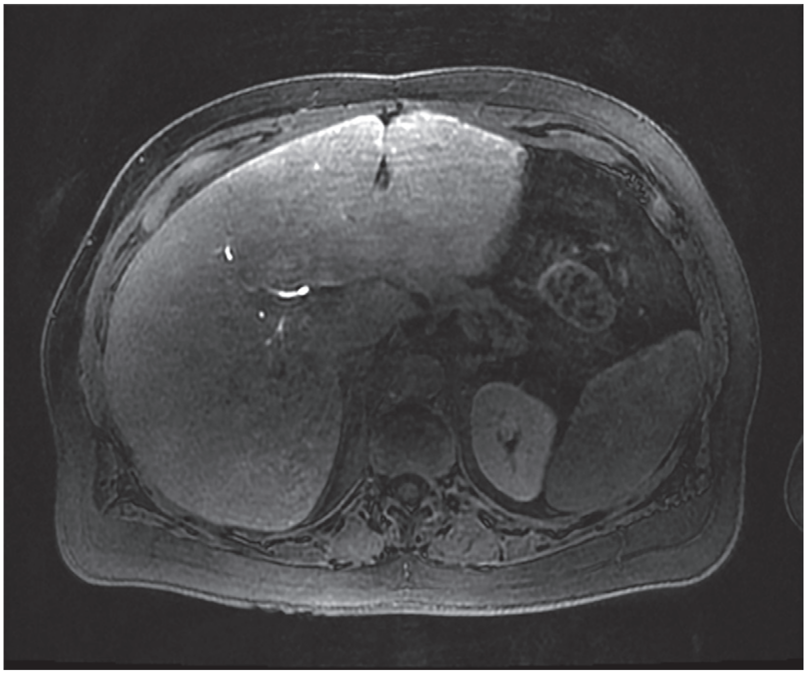

Hepatobiliary (HB) contrast agents are increasingly valuable diagnostic tools in MRI, offering a wider range of applications as their clinical use expands. Normal hepatocytes take up HB contrast agents, which are subsequently excreted in bile. This property creates a distinct HB phase providing valuable insights into liver function and biliary anatomy. HB contrast agents can assist in diagnosing a broad spectrum of HB diseases ranging from diffuse liver disease to focal hepatic lesions and can delineate anatomic details of the biliary tree. Understanding the pharmacodynamics of HB contrast agents is paramount to their appropriate clinical application and troubleshooting. HB phase hypointensity can arise from various diffuse and focal abnormalities that may or may not be associated with biliary excretion. Hypointensity during the HB phase can be broadly grouped into diffuse hypointensity, regional hypointensity, and focal lesions for better evaluation of the underlying cause. Abnormalities may arise from hepatic parenchymal, biliary, or vascular causes, or a combination thereof in each of the broad groups. Recognition of a suboptimal hypointense HB phase is important in the evaluation of focal lesions in patients with cirrhosis of the liver and particularly in those with hepatocellular carcinoma. Furthermore, hypointensity can also suggest the aggressiveness of malignancies such as hepatocellular carcinoma or colorectal metastases, which may affect the prognosis. It is essential to consider all imaging findings relative to the clinical context and the complete set of the MRI sequences performed for diagnosis of liver abnormalities. This comprehensive approach minimizes the risk of misinterpretation or pitfalls. The authors aim to equip radiologists with key insights for accurately understanding hypointensity in the HB phase, ultimately leading to more accurate diagnoses.

Introduction

Hepatobiliary (HB) contrast agents are taken up by functioning hepatocytes and excreted in the bile, and their paramagnetic properties cause shortening of the longitudinal relaxation time (T1) of the liver and the biliary tree (1). HB contrast agents cause marked and sustained enhancement of the liver parenchyma at T1-weighted MRI. The normal hepatic parenchyma shows initial hyperintensity due to hepatocellular uptake and then demonstrates slow excretion of HB contrast material into the bile (2). The HB phase occurs when an HB contrast agent reaches its peak concentration in the liver parenchyma, with concurrent excretion of contrast agent into the biliary system. An *adequate HB phase*, as defined by the American College of Radiology Liver Imaging Reporting and Data System (LI-RADS), does not require biliary excretion (2,3). Currently used HB contrast agents are gadolinium-based agents, namely gadoxetate and gadobenate. HB contrast agents are specifically used to improve lesion detection, characterize focal lesions, and assess the anatomic structure and function of the liver and biliary tree.

The initial clinical application of HB contrast agents was focused on uptake of the agents by normal hepatocytes, their biliary excretion, and the characterization of focal nodular hyperplasia (FNH). However, clinical experience has further demonstrated that hypointensity during the HB phase offers several additional valuable applications including detection of lesions, characterization of focal hepatic masses, Supplemental Material TestYour Knowledge CME

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Abbreviations: FNH = focal nodular hyperplasia, HB = hepatobiliary, HCC = hepatocellular carcinoma, LI-RADS = Liver Imaging Reporting and Data System, OATP = organic anion transport protein, PSVD = portosinusoidal vascular disorder, SOS = sinusoidal obstruction syndrome

TEACHING POINTS

- Generally, the severity of steatosis correlates with the degree of HB phase hypointensity, and this hypointensity can be heterogeneous, reflecting the heterogeneity of hepatic steatosis distribution.
- Paramagnetic iron shortens T2* relaxation, causing weaker signal intensity despite HB contrast agent uptake, with hypointensity worsening as iron overload severity increases.
- In patients with chronic liver diseases, diffuse hypointensity, which can be heterogeneous, is observed during the HB phase and may be associated with delayed, poor, or absent biliary excretion.
- A peripheral and reticular hypointensity pattern on HB phase MR images, with normal hepatic veins and inferior vena cava, is highly specific for SOS in a patient undergoing platinum-based chemotherapy and can sometimes be the sole imaging indicator.
- Although regional hypointensities on HB phase MR images are typically wedge shaped, corresponding to vascular supply or bile duct drainage, some diseases can affect both, resulting in irregular or large hypointense regions.

assessment of liver function, and determination of a suboptimal HB phase for characterization of hepatocellular carcinoma (HCC) (3–6). Although specific applications of HB contrast agents such as for detection of colorectal cancer metastases, exclusion of biliary leaks, and demonstration of biliary tree anatomy for liver donation have become routine, the findings during the HB phase are substantially broader.

This article delves into diffuse, regional, and focal hypointensity during the HB phase, exploring their various causes, characteristic appearances categorized as diffuse and focal, potential interpretation challenges associated with these findings, and the technical limitations that can lead to misinterpretation.

HB Contrast Agents and the Normal HB Phase

Current HB phase MRI primarily involves the use of two contrast agents: gadoxetate (Eovist, Primovist; Bayer Healthcare) and gadobenate (MultiHance; Bracco Diagnostics). Gadoxetate and gadobenate are both ionic contrast agents with a linear molecular structure, sharing a common core structure of gadolinium-DTPA, with the addition of benzyl groups to promote selective hepatocellular uptake, distinguishing them from traditional gadolinium chelates (7). Similar to extracellular contrast agents, HB contrast agents are first distributed in the extracellular fluid, allowing dynamic studies, then are taken up by hepatocytes and excreted into bile during the HB phase (4,6,8). Both gadoxetate and gadobenate are transported into the hepatocytes by means of organic anion transport proteins (OATP) 1B1 and 1B3 and then excreted into the biliary canaliculi by means of multidrug resistance–associated protein 2 (3,9). Multidrug resistance-associated protein 3 on the sinusoidal surface membrane of the hepatocyte mediates excretion back into the sinusoidal space (3). While both HB contrast agents are taken up by hepatocytes and excreted in bile, gadoxetate offers practical advantages. Its higher biliary excretion (50% versus 5%) leads to quicker peak enhancement (20 minutes vs 1–3 hours) and shorter half-life (2 hours vs 4–6 hours) (7), translating to an earlier and higher signal intensity HB phase. The HB contrast agent properties are summarized in Table 1.

Manganese-based agents have been investigated as alternatives to gadolinium-based agents for the HB phase. Mangafodipir trisodium was previously approved for clinical use; however, due to poor clinical performance and concerns over toxicity, it was withdrawn from the U.S. market in 2003 and from the European Union in 2010. Orally administered manganese chloride is under clinical trial but is not currently U.S. Food and Drug Administration–approved for imaging (10).

HB phase MR images are usually acquired during a T1-weighted fat-suppressed MRI sequence. A normal HB phase demonstrates homogeneous hepatic parenchymal hyperintensity (Fig 1). The increased signal intensity occurs because of transient interaction of HB contrast agents with hepatocellular proteins, increasing the T1-relaxivity of the liver by a factor of four to six (7). The parenchyma is hyperintense to the hepatic vasculature, spleen, and renal parenchyma. The biliary excretion of contrast material renders the bile ducts markedly hyperintense and with higher signal intensity than that of the background liver. In addition, there is excretion of contrast material by the kidneys. The proportion of excretion of gadoxetate into biliary and renal routes is dependent on liver and renal function in an individual patient (11). Biliary excretion, while a defining feature of the HB phase, does not guarantee normal liver function. Biliary excretion can occur even with compromised hepatocyte function, although it is delayed and diminished (3). The liver parenchymal enhancement in the HB phase depends on several factors including the presence of hepatocytes, vascular supply, the expression and function of both OATPB and multidrug resistance–associated protein transporters, biliary duct patency, the contribution of the interstitial compartment, and the kidney clearance rate (2,3,12,13).

Having a frame of reference of a normal HB phase is particularly important when assessing for an optimal HB phase (Fig 2). The normal hepatic parenchyma should be hyperintense to (ie, higher signal intensity) the intrahepatic blood vessels, kidney, and spleen (3,14,15). An HB phase is considered suboptimal under the American College of Radiology LI-RADS criteria if the hepatic parenchyma is not clearly hyperintense relative to intrahepatic blood vessels and precludes assessment of hepatic observations (3). Visual evaluation of the signal intensity of the liver relative to that of the portal vein or the kidneys is useful for assessment of HB phase adequacy (14,15). The HB phase may vary in signal intensity among normal individuals due to differences in elimination of HB contrast agents from hepatocytes and genetic polymorphisms in expression of OATP1B1 receptors (7,16). Changes in the HB phase during follow-up are useful for assessment of changes in liver function and progression of disease.

Table 1: Contrast Agents for the HB Phase and Their Properties

Property	Gadobenate Dimeglumine	Gadoxetate Disodium
Paramagnetic substance	Gadolinium	Gadolinium
Typical dosage	0.05–0.1 mmol/kg 0.1–0.2 mL/kg	0.025–0.05 mmol/kg 0.1–0.2 mL/kg
Route	Intravenous	Intravenous
Liver uptake (%)	3–5	50
Hepatocyte uptake mechanism	OATP1B1, OATP1B3	OATP1B1, OATP1B3
Uptake start (min)	30	Approximately 1
HB phase peak (min)	40–120	10–20
HB phase duration (min)	240–360	120
Excretion	Predominantly renal, 3%–5% biliary	Approximately 50% renal excretion, approximately 50% biliary

Note.— Gadobenate dimeglumine (MultiHance; Bracco Diagnostics); gadoxetate disodium (Eovist; Bayer Healthcare).

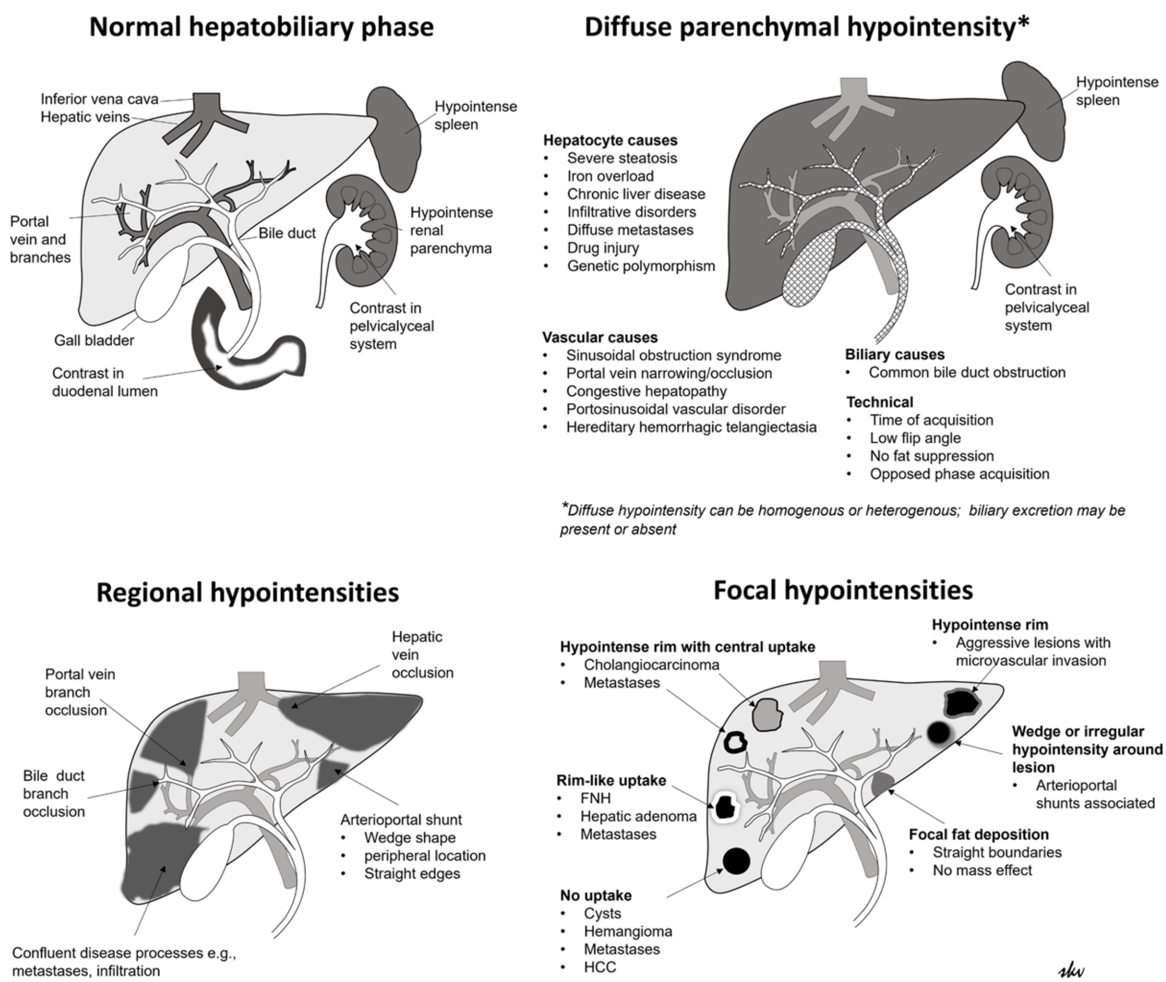

Figure 1. Illustrations show normal HB phase imaging and diffuse, regional, and focal hypointensities.

Hypointensity during the HB Phase

Causes of hypointense findings are broad and can be categorized into those that are hepatic parenchymal in origin and those that are focal lesions (Tables 2, 3). Parenchymal hypointense observations are further subdivided into diffuse, regional (involving one or more liver segments), or focal patterns (Fig 1).

Diffuse Hypointensity

Diffuse hypointensity during the HB phase occurs when the entire liver parenchyma exhibits a lower signal intensity than that of the normal liver parenchyma. This manifests as liver parenchyma that is not clearly hyperintense to the intrahepatic vessels, spleen, and kidneys. Identifying causes of

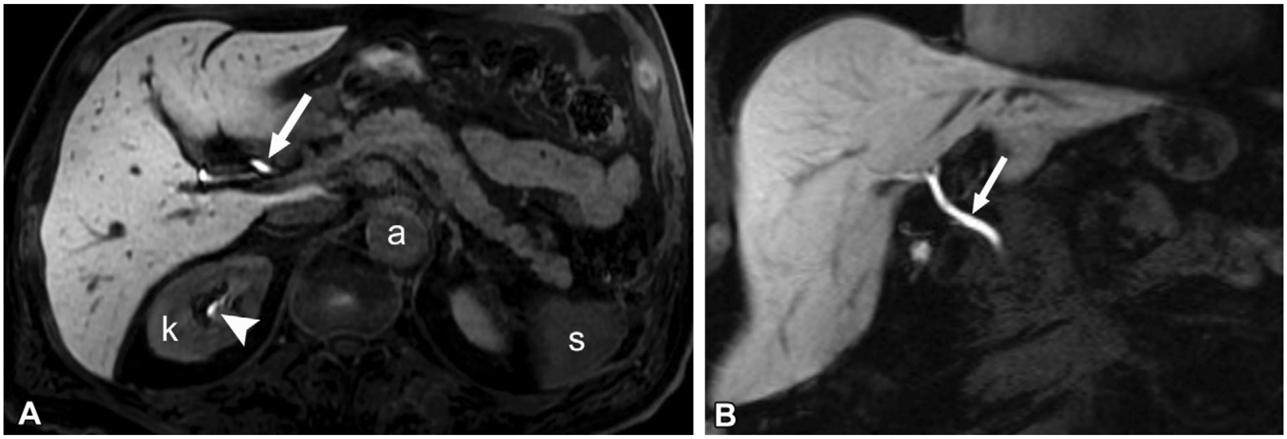

Figure 2. Normal HB phase MR images in a 50-year-old man. Axial **(A)** and coronal **(B)** images obtained at 20 minutes and 21 minutes after injection of gadoxetate show normal uptake in the liver parenchyma. The hepatic parenchyma is homogeneous and hyperintense to the intrahepatic vessels, the aorta *(a)*, kidneys *(k)*, and spleen *(s)*. The most hyperintense structures are the biliary tree (arrow). Note the normal renal excretion on the axial image (arrowhead in **A**).

diffuse parenchymal hypointensity is important due to its prevalence and the increasing global burden of chronic liver diseases (17). Accurately identifying diffuse hypointensity can be challenging, especially when there is normal biliary excretion. Diffuse hypointensity is qualitative, with less than expected signal intensity of the hepatic parenchyma, without a signal intensity cutoff or specific ratio. Furthermore, technical parameters can lead to misinterpretations, including both false-positive and false-negative results.

Hepatocytic-related Hypointensity.—Diffuse hepatic steatosis such as that seen in patients with metabolic dysfunction–associated fatty liver disease is a common clinical finding. Other causes of steatosis include alcohol use disorder, drugs or chemotherapy, and metabolic disorders such as glycogen storage disease. A normal appearance is usually seen during the HB phase in patients with simple mild to moderate steatosis. Diffuse HB phase hypointensity in patients with mild to moderate hepatic steatosis is suggestive of underlying hepatocyte dysfunction or poor liver function secondary to inflammation and/or fibrosis, and this finding can be used to differentiate simple steatosis from steatohepatitis (18). However, severe hepatic steatosis may manifest by itself as diffuse HB phase hypointensity (Fig 3). Probable reasons include decreased retention of HB contrast agent in hepatocytes due to fat vacuoles, ballooned hepatocytes compromising the hepatic sinusoidal space, lobular inflammation, and associated fibrosis in advanced stages (18). In addition, loss of signal intensity may be due to fat-water signal intensity interference on out-of-phase MR images or from routine fat suppression (3). Generally, the severity of steatosis correlates with the degree of HB phase hypointensity, and this hypointensity can be heterogeneous, reflecting the heterogeneity of hepatic steatosis distribution (19). In these patients, enhancement of the bile ducts shows the uptake of HB contrast media and excretion consistent with preserved hepatocytic function (Fig 3). Delayed or absent biliary excretion may indicate acute injury (eg, drug-induced liver injury) or advanced chronic liver disease.

Hepatic iron overload causes diffuse HB phase hypointensity (Fig 4). Paramagnetic iron shortens T2* relaxation, causing weaker signal intensity despite HB contrast agent uptake, with hypointensity worsening as iron overload severity increases. Enhancement of bile ducts confirms uptake of HB contrast media, and excretion is suggestive of preserved function. However, when there is delayed or absent excretion of contrast material in bile ducts, the patient should be further examined for underlying chronic liver disease. The presence of iron overload may be confirmed by observing the loss of signal intensity on in-phase gradient-recalled-echo MR images in comparison with out-of-phase MR images, or with T2* or R2* maps to quantify liver iron content to help identify iron overload as a cause of diffuse hypointensity.

Chronic liver disease may be associated with hepatic dysfunction. In patients with chronic liver diseases, diffuse hypointensity (Figs 5, 6), which can be heterogeneous, is observed during the HB phase and may be associated with delayed, poor, or absent biliary excretion (5). Patients with cirrhosis have impaired hepatic function, resulting in reduced biliary excretion and a compensatory increase in renal excretion and a prolonged plasma half-life of gadoxetate. This leads to decreased contrast between the liver parenchyma and the intrahepatic vessels, resulting in a suboptimal HB phase (20). Patients with marked hepatic impairment show global hypointensity, particularly if there is an elevated bilirubin level (21,22). A general cutoff of greater than 3 mg/dL (51.31 µmol/L) in patients is used to determine whether a study will show adequate uptake and provide diagnostic images (21). Furthermore, in patients with diffuse fibrosis in addition to loss of hepatocytes, deposition of collagen around the sinusoids and hepatocytes also reduces access to receptors for HB contrast agent uptake (20) resulting in hypointensity in the regions of fibrosis. Conventional MRI sequences typically show imaging manifestations of chronic liver disease (Figs 5, 6).

Hypointensity during the HB phase may indicate poor liver function in patients with chronic liver disease. The relative liver enhancement ratio, calculated from the difference between enhancement during the HB phase and that on

Table 2: Categorizations of Parenchymal Hypointensities and Hypointense Lesions

Diffuse hypointensities
Hepatocyte-related
Hepatic steatosis
Iron overload
Chronic liver disease: fibrosis or cirrhosis
Infiltrative disorders: amyloidosis and metastases
Drug injury
Genetic polymorphism
Vascular
Congestive hepatopathy
Portosinusoidal vascular disorder (PSVD)
Hereditary hemorrhagic telangiectasia
Acute portal vein narrowing or occlusion
Biliary: common bile duct obstruction
Other: technical
Timing
Opposed phase
No fat suppression
Low flip angle
Regional hypointensities
Hepatocyte-related
Stereotactic body radiation therapy
Percutaneous thermal ablation
Radioembolization
Vascular
Acute portal vein branch occlusion
Portal vein embolization
Hepatic vein occlusion
Hepatic artery occlusion
Biliary: bile duct branch occlusion
Other
Focal hypointensities
Hepatocyte-related focal fat deposition
Vascular arterioportal shunts
Bile ducts: cholangitis
Other: focal confluent fibrosis

Table 3: Hypointense Lesions

Hepatocellular lesions
Benign: hepatocellular adenoma
Malignant: HCC
Nonhepatocellular lesions
Benign
Simple benign cysts
Hemangiomas
Abscess
Malignant
Cholangiocarcinoma
Metastases
Other malignancies

noncontrast T1-weighted MR images, can allow prediction of the stage of liver fibrosis and is associated with long-term patient outcomes (23). The relative liver enhancement also correlates with liver stiffness measured with MR elastography (23). A simple semiquantitative scoring system during the HB phase, the functional liver imaging score, was proposed to stratify patients, identifying those at risk for decompensation and mortality (7). The functional liver imaging score is based on qualitative assessment of three parameters, and scores are added (Fig S1). The scores range from 0 to 6, with a low functional liver imaging score associated with a high mortality rate.

Infiltrative disorders such as diffuse infiltrative metastatic disease can cause widespread HB phase hypointensity due to extensive replacement of healthy liver tissue (Fig 7). Often there is a known history of cancer, progression over time, or metastases elsewhere to assist in the diagnosis. Uncommonly, hepatic amyloidosis can lead to diffuse hypointensity (24). HB phase hypointensity tends to be heterogeneous in patients with these conditions. Some imaging findings to assist in making the diagnosis of amyloidosis include hepatomegaly, markedly elevated liver stiffness without findings of cirrhosis, and poor enhancement with extracellular contrast agents (25,26).

Drug-induced liver injuries are complex and diverse and range from mild to fulminant liver failure. Drug-induced liver injuries can be caused by various medications and supplements. Some injuries are predictable and dose-dependent toxicities, while others are idiosyncratic and patient dependent (27). Drug-induced liver injuries can manifest as hepatocellular injury, cholestatic injury, or mixed injuries (28). Diffuse HB phase hypointensity may be a result of mixed processes of hepatocellular injury and cholestatic injury. Diagnosis of drug-induced liver injuries typically requires clinical correlation for temporality of illness with drug administration and withdrawal of the offending agent to establish a diagnosis (28). Sinusoidal obstruction syndrome (SOS) is a form of drug-induced liver injury and is discussed under "Vascular Causes of Hypointensity."

Genetic polymorphisms can result in variable expression of OATP1 receptors, resulting in decreased HB contrast agent uptake into hepatocytes (16). This variability could be erroneously interpreted as decreased function in the absence of other visible factors such as diffuse hepatic steatosis or iron overload. Furthermore, there are studies demonstrating decreased hepatocytic uptake in patients taking over-the-counter medications such as nonsteroidal anti-inflammatory drugs, suggesting not only interindividual variability but also intraindividual variability (29,30). Diagnosis of the drug interaction and genetic polymorphism can be difficult and may be proposed when a patient shows normal biliary excretion but has no known chronic liver disease and no technical causes of hypointensity during the HB phase.

Vascular Causes of Hypointensity.—Vascular abnormalities can result in diffuse HB phase hypointensity (Figs 8, S2). Vascular causes include congestive hepatopathy, portosinusoidal vascular disorder (PSVD), hereditary hemorrhagic telangiectasia, SOS, arteriovenous malformations, and recent portal vein narrowing or occlusion. Vascular causes of diffuse HB

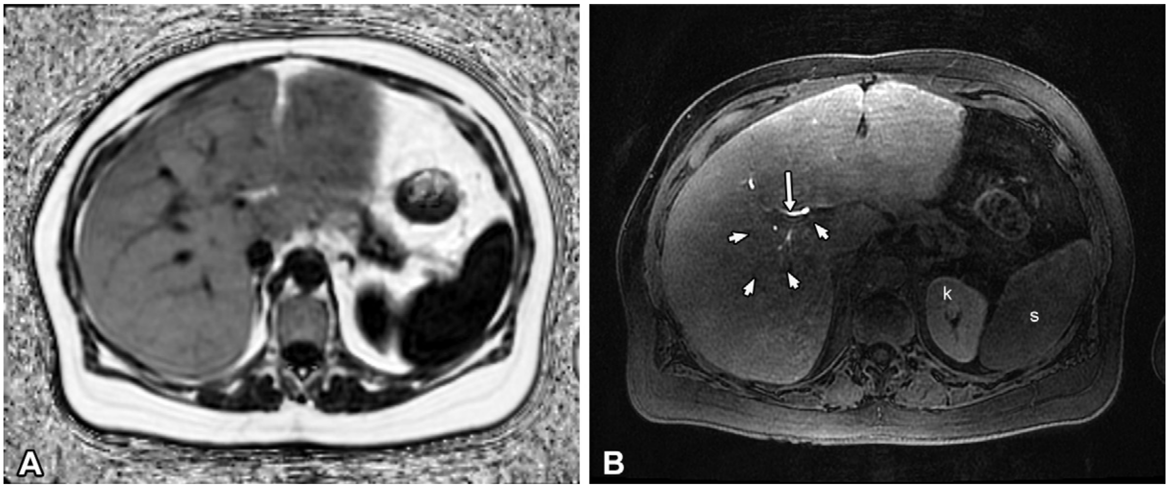

Figure 3. Hepatic steatosis in a 45-year-old woman with a history of pancreatic resection for a neuroendocrine tumor, no history of chemotherapy, and normal serum liver function test results. **(A)** Axial proton density fat fraction map shows a mean proton density fat fraction of 45% in the right lobe and 32% in the left lobe. **(B)** Axial gadoxetate-enhanced HB phase MR image shows diffuse hypointensity and normal biliary excretion (arrow in **B**). Note that the intrahepatic vessels are not well depicted as hypointense structures (arrowheads in **B**) as they are on the proton density fat fraction map. The liver parenchyma is only mildly hyperintense to the spleen *(s)* and not hyperintense to the kidney *(k)*. The left lobe and the anterior right lobe of the liver show higher signal intensity than that of the posterior right lobe due to artifact from the surface receiver coil and less fat deposition in the left lobe than in the right lobe. Liver biopsy results showed moderate (>60% hepatocytes) hepatic steatosis, without any evidence of steatohepatitis or fibrosis.

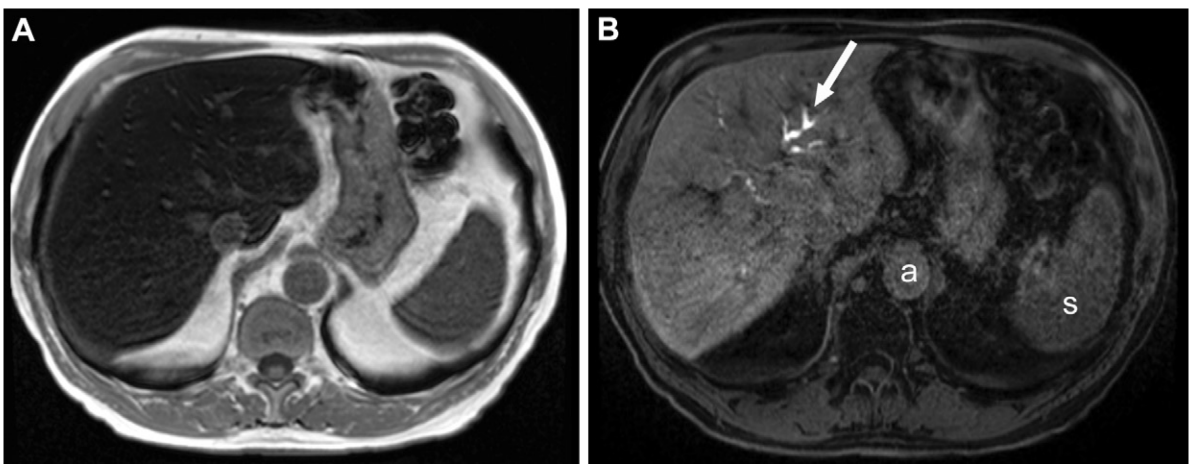

Figure 4. Diffuse HB phase hypointensity in a 67-year-old man with diffuse iron overload secondary to hemochromatosis. **(A)** Axial in-phase gradient-recalled-echo T1-weighted MR image shows the liver and spleen as hypointense, suggesting the presence of hepatic and splenic iron overload. **(B)** Axial HB phase gadoxetate-enhanced MR image shows diffusely decreased hepatic parenchymal signal intensity that is not unequivocally hyperintense to the intrahepatic vessels, adjacent aorta *(a)*, spleen *(s)*, and kidneys (not shown) despite biliary excretion (arrow in **B**).

phase hypointensity tend to have some common imaging features including heterogeneous arterial phase enhancement but homogeneous or nearly homogeneous enhancement during the portal venous phase, with the exception of congestive hepatopathy, arteriovenous malformations, and PSVD. Despite the homogeneous portal venous phase, the HB phase usually has a heterogeneous appearance. The appearance of the diffuse hypointensity may suggest a cause: a peripheral reticular pattern in SOS, a geographical pattern in PSVD, a patchy pattern in SOS and arteriovenous malformations, and peripheral hypointensity in congestive hepatopathy (31–33). The liver function is maintained until the end stage of these vascular diseases, when fibrosis develops, which can progress to cirrhosis. SOS is associated with portal hypertension and splenomegaly as well as the development of FNH-like lesions in later stages.

A vascular cause of hypointensity can typically be elucidated from conventional MRI images and the patient's clinical history, with the exception of PSVD, for which liver biopsy is fundamental for diagnosis (34,35). PSVD is an umbrella term that encompasses a group of vascular diseases affecting intrahepatic portal venules and sinusoids regardless of the presence of portal hypertension. PSVD was previously considered to be in the noncirrhotic portal hypertension group

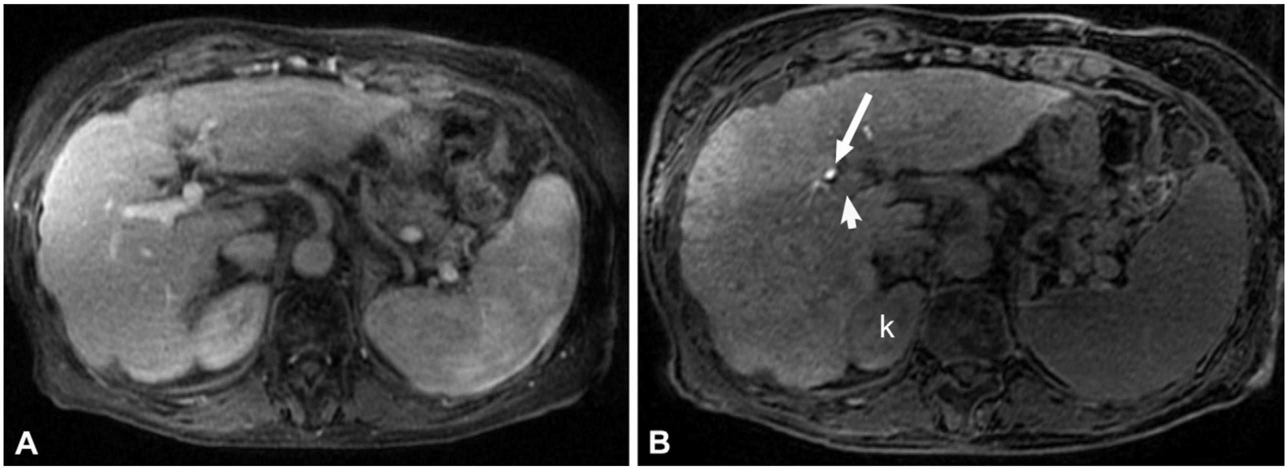

Figure 5. Diffuse HB phase hypointensity in a 75-year-old woman with cirrhosis secondary to metabolic dysfunction–associated steatotic liver disease. **(A)** Portal venous phase MR image shows surface nodularity and a lobulated outline of the liver consistent with cirrhosis, as well as sequela of portal hypertension with splenomegaly. **(B)** Axial HB phase gadoxetate-enhanced MR image at 20 minutes shows that the liver is diffusely hypointense and not unequivocally hyperintense to the portal vein (arrowhead) and kidney *(k)*. However, there is biliary excretion (arrow). This image is considered suboptimal for LI-RADS categorization because the liver is not unequivocally hyperintense to the portal vein.

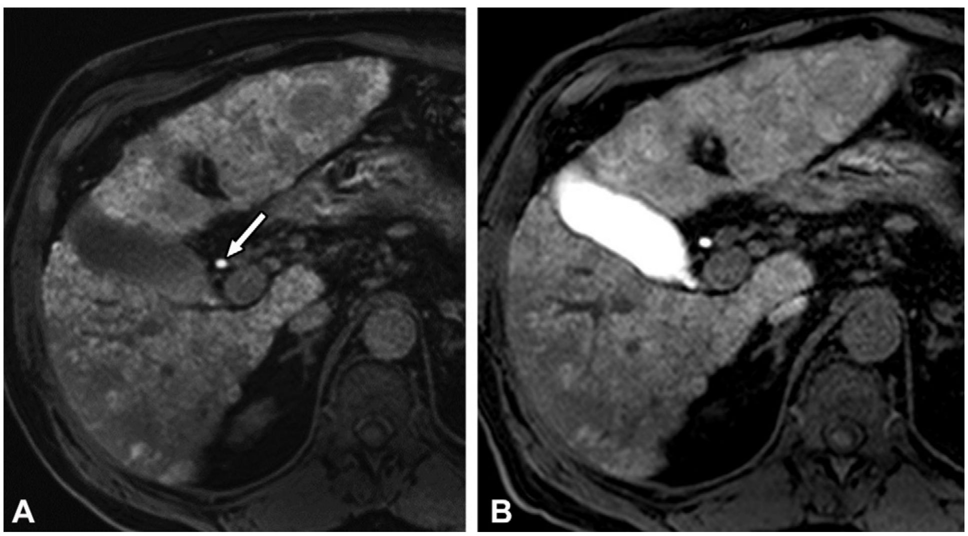

Figure 6. Diffuse nodular HB phase hypointensity in a 53-year-old man due to cirrhosis from hepatitis B virus. **(A)** Axial HB phase MR image acquired 20 minutes after administration of gadoxetate shows a diffusely and heterogeneously hypointense liver parenchyma despite excretion of contrast material into the biliary tree, as evidenced by the hyperintense common bile duct (arrow). **(B)** Axial MR image acquired with a prolonged wait of 120 minutes shows increased parenchymal signal intensity and filling of the gallbladder and common bile duct with contrast material.

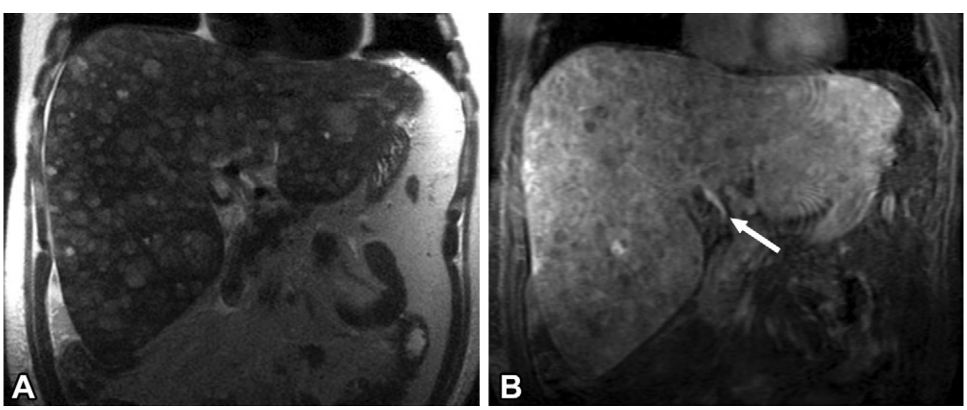

Figure 7. Diffuse nodular thyroid carcinoma metastases in a 73-year-old woman. Coronal T2-weighted **(A)** and coronal HB phase gadoxetate-enhanced **(B)** MR images show diffuse and confluent metastases that are heterogeneously T2 hyperintense throughout the liver. There is also diffuse heterogeneous signal hypointensity (arrow in **B**). There is biliary excretion, which is suggestive of preserved liver excretory function.

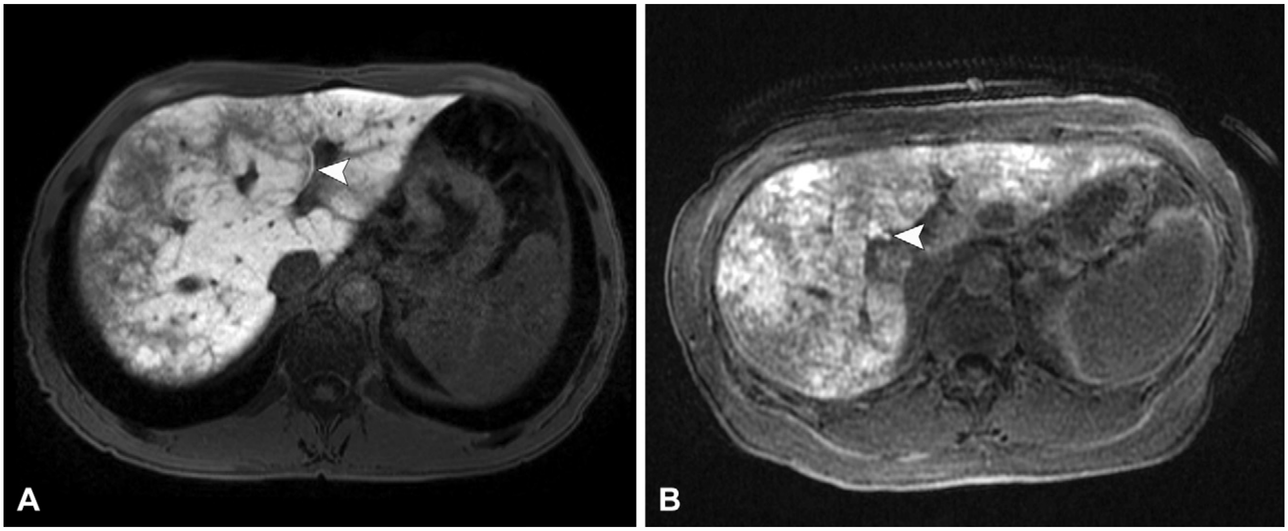

Figure 8. SOS causing diffuse and heterogeneous hypointensity on HB phase images in two patients. **(A)** Axial HB phase MR image in a 56-year-old man with colorectal carcinoma who underwent oxaliplatin-based chemotherapy shows diffuse reticular hypointensities predominantly in the periphery of the liver. Before chemotherapy, the HB phase images showed normal enhancement (not shown). **(B)** Axial HB phase MR image in a 51-year-old woman with colon cancer after she underwent oxaliplatin-based chemotherapy shows extensive reticular hypointensities throughout the liver. Before initiation of chemotherapy, the liver parenchyma showed normal HB phase enhancement (not shown). Note that both images show biliary excretion (arrowhead) in addition to mild splenic enlargement.

of diseases. Imaging findings of PSVD (Fig S2) can include a lack of findings of cirrhosis (eg, surface nodularity), hypertrophy of segments IV and I with peripheral atrophy, normal to slightly elevated liver stiffness, findings of portal hypertension, and FNH-like lesions (29). PSVD shows diffuse geographic HB phase hypointensity.

SOS is a well-recognized entity that results from injury to the liver sinusoidal endothelial cells. A clinical history of having undergone a hematopoietic stem cell transplant, chemotherapy with platinum-based agents, ingestion of pyrrolizidine alkaloids, and radiation therapy are important for diagnosis (36). MRI remains the imaging modality of choice for SOS, with the added benefit of assessment of treatment response in any underlying liver tumors. A peripheral and reticular hypointensity pattern on HB phase MR images (Fig 8), with normal hepatic veins and inferior vena cava, is highly specific for SOS in a patient undergoing platinum-based chemotherapy and can sometimes be the sole imaging indicator (33,37). Other imaging findings may include hepatomegaly with or without heterogeneous enhancement, an increase in the size of the spleen or splenomegaly, and findings of portal hypertension. Subtle parenchymal abnormalities on T2-weighted, diffusion weighted, and contrast-enhanced T1-weighted MR images may also be evident (33).

Congestive hepatopathy occurs in patients with a variety of conditions from ischemic cardiomyopathy to congenital heart disease including that which occurs after Fontan surgery. On HB phase MR images, a reticular pattern of hypointensity appears predominantly in the periphery and in the regions of congestion representing areas of dilated sinusoids (38). This pattern of hypointensity may appear similar to SOS; however, other signs of congestion including dilated hepatic veins and a dilated inferior vena cava are usually not present in patients with SOS (33). The HB phase hypointensity may be accentuated in areas of fibrosis, if present. FNH-like lesions are common, but HCCs rarely develop in this patient population (32). Supportive imaging findings for a diagnosis of congestive hepatopathy include a dilated inferior vena cava and hepatic veins, contrast agent reflux into hepatic veins during the arterial phase, a peripheral heterogeneous pattern of hepatic enhancement in earlier phases, and fibrosis in advanced cases.

Recent portal vein narrowing or occlusion may result in diffuse HB phase hypointensity, while chronic occlusion typically results in cavernous transformation and at least partial normalization of portal vein blood flow. The uptake of the HB contrast agent by hepatocytes is related to the maintenance of portal flow, although the complex relationship between portal flow and HB contrast agent uptake is still not well understood and is thought to be regulated by OATPB receptor expression and function (39).

Hereditary hemorrhagic telangiectasia (ie, Osler-Weber-Rendu syndrome) is a rare autosomal-dominant condition and is characterized by arteriovenous malformations throughout the body, with hepatic involvement typically seen as diffuse heterogeneous involvement rather than as discrete arteriovenous malformations. Complex vascular malformations can develop over time from telangiectases, with arteriosystemic or arterioportal shunts (40). Abnormal blood flow and physiologic characteristics result in dilated hepatic veins, heterogeneous enhancement on arterial and portal phase MR images, and diffuse HB phase hypointensities. FNH-like lesions may also develop throughout the liver. Over time, due to altered hemodynamics, patients may develop imaging findings of high-output cardiac failure and portal hypertension (40).

Biliary Causes of Hypointensity.—Diffuse HB phase hypointensity can occur with complete common bile duct occlusion and often without excretion of contrast material into the

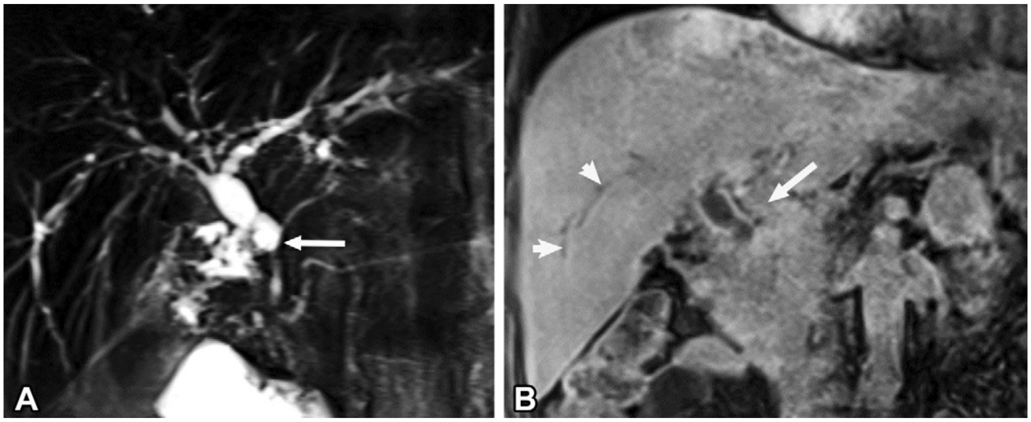

Figure 9. Primary sclerosing cholangitis with a high-grade common bile duct stricture in a 46-year-old man. **(A)** Coronal MR cholangiopancreatographic image shows a common bile duct stricture (arrow). There was normal parenchymal liver enhancement on arterial and portal venous phase MR images (not shown) but diffuse hypointensity on gadoxetate-enhanced HB phase MR images and no biliary excretion. **(B)** Coronal HB phase MR image shows that the liver parenchyma is not hyperintense to the portal vein (arrow), and the intrahepatic vessels, which are typically hypointense during the HB phase, are not visualized. However, dilated right-sided bile ducts are visualized as hypointense structures (arrowheads) because there is no excretion of gadoxetate.

biliary system. However, isolated ductal occlusion at lobar and segmental ductal occlusions do not cause diffuse hypointensity, but rather regional or segmental hypointensity. The liver parenchyma usually shows normal dynamic contrast enhancement if the hepatic artery and portal vein are patent and uninvolved by the disease process. Due to cholestasis, there is decreased hepatocyte uptake in the acute phase secondary to competition by bilirubin for the OATP receptors and background inflammation (41). In chronic disease processes, there may be associated biliary fibrosis or underlying chronic hepatic dysfunction resulting in the diffuse hypointensity (Fig 9).

Technical Causes of Diffuse Hypointensity.—It is important to identify technical reasons for diffuse hypointensity to avoid misinterpretation as a hepatocyte disorder. Early HB phase acquisition before peak enhancement can cause false hypointensity. Optimal timing for HB phase acquisition is relative to liver function; some patients may require a delay for maximized uptake (Fig 6). Flip angles also influence the appearance of the liver (Fig 10), with increased flip angles (25°–45°) improving the contrast-to-noise ratio and the HB phase appearance (2,4,42). The appearance of the liver parenchyma is relative to the signal intensity of the surrounding tissues during the HB phase. For example, fat suppression in the HB phase is important to suppress abdominal fat signal intensity (Fig 11) because the fat can appear with higher signal intensity than that of the liver and give a false impression of HB phase hypointensity. Use of an HB contrast agent with an extracellular contrast agent in a dual-contrast protocol (43) can result in a suboptimal HB phase due to circulating extracellular contrast material having a longer half-life in the vascular space than that of HB contrast material, resulting in hyperintense vessels (Fig S3).

Regional Hypointensities

Regional hypointensities affect multiple hepatic segments or the entire lobe and are often caused by occlusions of branches of portal or hepatic veins or major bile ducts. Although regional hypointensities on HB phase MR images are typically wedge shaped, corresponding to vascular supply or bile duct drainage, some diseases can affect both, resulting in irregular or large hypointense regions.

Hepatocytic-related Hypointensities.—Regional hypointensities during the HB phase are common after the patient undergoes locoregional therapy for liver tumors. These therapies damage hepatocytes through various mechanisms, causing regional hypointensity. Posttreatment changes after stereotactic body radiation therapy evolve over time. Although the epicenter of the radiation is targeted at the specified lesion and the dose decrease is rapid in stereotactic body radiation therapy, a substantial radiation dose reaches the surrounding parenchyma, resulting in a focal liver reaction (44). Typically, hypointensity is seen around the treated tumor (Fig 12). Several factors may contribute to this hypointensity, including damage to Kupffer and sinusoidal endothelial cells, activated inflammatory pathways, and altered hepatocytic signaling (45). Radiation-induced liver injury may be underestimated on images from conventional MRI sequences, but it is well visualized as a region of hypointensity during the HB phase (46).

Radioembolization delivers targeted radiation to the targeted tumor and nontargeted radiation to the adjacent liver parenchyma (47). The radiation injury occurs typically in the parenchyma and is supplied by the hepatic artery branches that are embolized for tumor treatment. On images from conventional MRI sequences, the targeted mass and surrounding liver may demonstrate posttreatment changes such as T2 signal hyperintensity, while HB phase signal hypointensity demonstrates the true extent of hepatic involvement, similar to that with stereotactic body radiation therapy (Fig 13). The regional hypointensity may not be well defined because the radiation injury may be extensive and may not be confined to one vascular territory due to collateral vessels and abnormal tumor vessels.

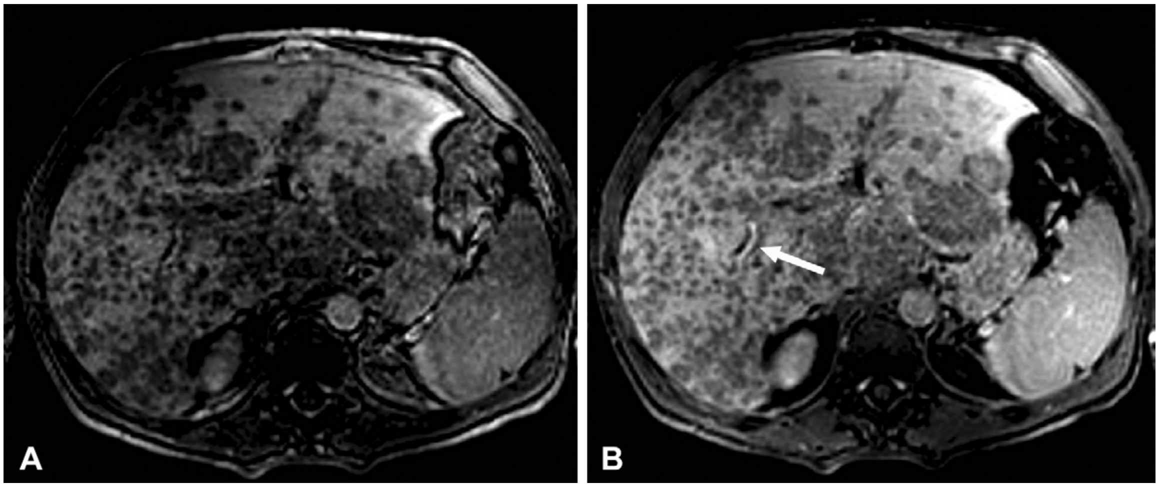

Figure 10. Alcoholic cirrhosis and diffuse HCC in a 47-year-old man. Axial HB phase MR images obtained at 60 minutes after injection of gadobenate with flip angles of 15° **(A)** and 25° **(B)** show an improved contrast-to-noise ratio, with the higher flip angle and improved differentiation of the hypointense nodules representing the HCC and the excretion of contrast material into the right intrahepatic duct (arrow in **B**). There is improved depiction of the burden of disease with the higher flip angle.

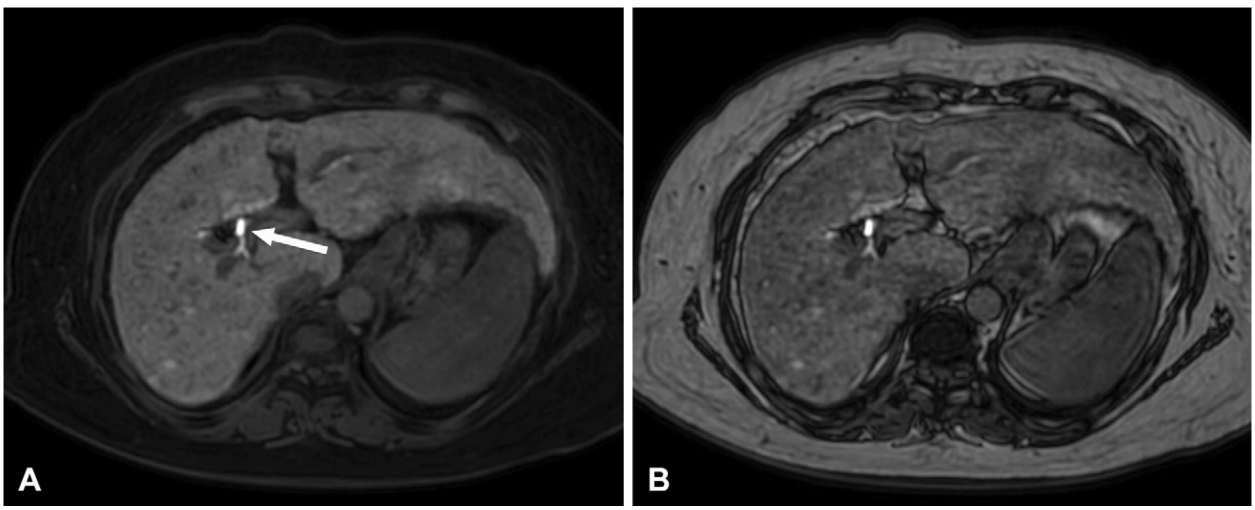

Figure 11. Biopsy-proven cirrhosis secondary to metabolic dysfunction–associated steatotic liver disease in a 46-year-old woman. Gadoxetate-enhanced HB phase Dixon-based T1-weighted water-only **(A)** and out-of-phase **(B)** MR images show the cirrhotic morphology of the liver with splenomegaly. The mean proton density fat fraction of the liver was 15%. Note the normal biliary excretion (arrow in **A**). The liver is heterogeneously hyperintense to the intrahepatic vessels and spleen in **A**, where the signal from abdominal subcutaneous fat and mesenteric fat is suppressed, whereas the liver parenchyma appears heterogeneously hypointense to the mesenteric fat and subcutaneous fat and only mildly hyperintense to the spleen in **B,** in which the signal from abdominal subcutaneous fat and mesenteric fat is not suppressed.

Percutaneous thermal ablation of liver tumors typically creates an ablation zone larger than that of the initial tumor. This zone encompasses the tumor and a rind of liver parenchyma to ensure negative margins. HB phase MR images demonstrate an area of hypointensity larger than that of the initial target (Fig 14). The ablation area is seen on images from conventional MRI sequences as T2 hyperintensity, T1 hypointensity to hyperintensity, and possible restricted diffusion, but with no internal enhancement if the treatment was successful. Careful viewing of images from the HB phase with the knowledge of treatment type and pre- and posttreatment MRI sequences are required to avoid misinterpreting the larger HB phase hypointensity as tumor growth. Typically, the ablation zone hypointensity has sharp borders and gradually reduces in size over months or years. An irregular outline, nodularity with enhancement in the dynamic phases, and/or restricted diffusion are suspicious features for local recurrence.

Imaging during the HB phase can be valuable in patients with chronic liver disease who are undergoing repeated or combination locoregional therapy. It helps to assess residual liver function and volume, aiding the planning of future treatments and preventing liver failure (4).

Confluent fibrosis can occur in patients with conditions such as primary biliary cholangitis and primary sclerosing cholangitis and may cause large regions of fibrosis that replace the normal liver parenchyma. These large confluent

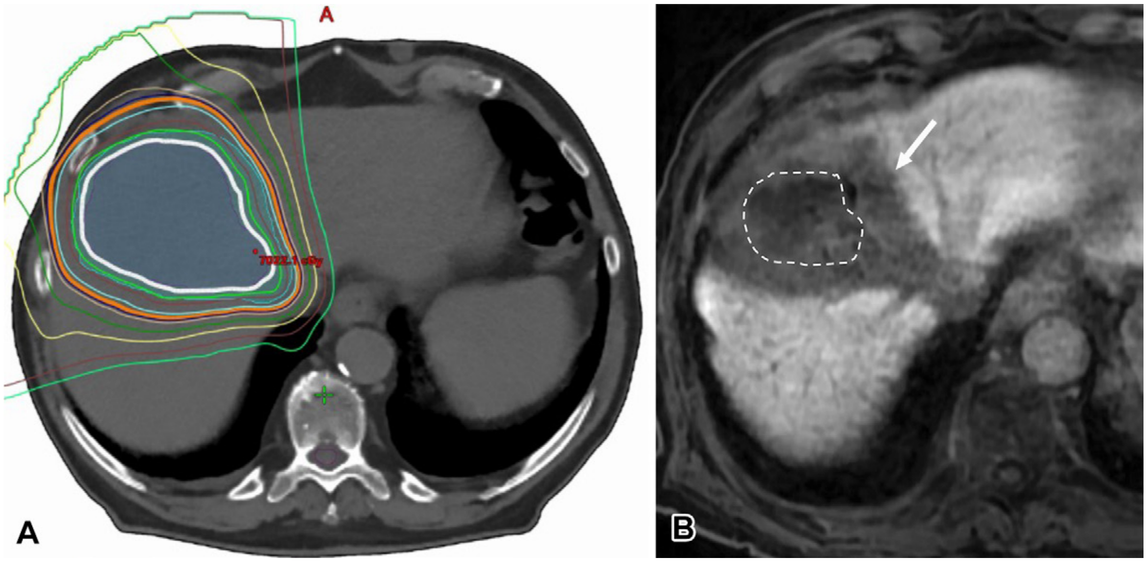

Figure 12. External beam stereotactic body radiation therapy for cholangiocarcinoma. **(A)** Axial treatment planning CT image shows the isoattenuating lines for radiation of the tumor. **(B)** Gadoxetate-enhanced HB phase MR image obtained 3 months after stereotactic body radiation therapy shows hypointensity (arrow) in the posttreatment irradiated liver encompassing the treated tumor (dotted outline), corresponding to the radiation planning zone. The area of hypointensity is larger than the treated tumor.

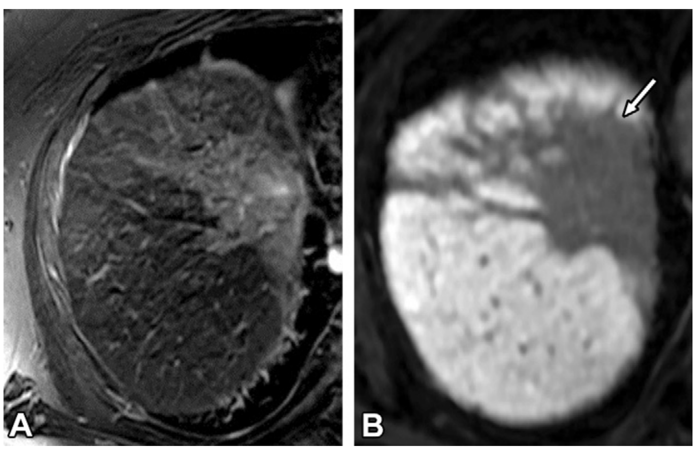

Figure 13. Radioembolization for cholangiocarcinoma. **(A)** Axial T2-weighted MR image acquired 10 months after treatment shows mild regional signal hyperintensity. **(B)** Corresponding axial gadoxetate-enhanced HB phase MR mage shows irregular hypointensity (arrow), which represents the irradiated tumor and adjacent liver parenchyma. The presence of residual tumor cannot be accurately assessed during the HB phase because the tumor and adjacent parenchyma are not distinguishable. Review of other MR images is necessary for evaluation of tumor response.

areas of fibrosis can manifest as irregular hypointense regions during the HB phase (Fig S4).

Vascular Causes of Hypointensity.—Various vascular conditions can cause regional hypointensities including arterioportal shunts, acute portal vein branch occlusion, portal vein embolization (Fig 15), hepatic vein occlusion, and hepatic artery occlusion. The distribution of the region of hypointensity correlates with that of the underlying vascular abnormality or occlusion.

Arterioportal shunts can cause HB phase hypointensity (Fig S5). These shunts may result from several causes and may be congenital or acquired. When they are detected, close assessment for an underlying lesion is required, particularly in a patient with known risk factors for metastases or HCC. Up to 15% of arterioportal shunts can be seen as HB phase hypointensity (48). These typically small, peripheral, wedge-shaped hypointensities match the perfusion change appreciated on arterial phase MR images and become isointense in later dynamic phases.

A portal vein thrombus, whether it is bland, malignant, or iatrogenic, alters portal venous flow and HB contrast agent uptake by hepatocytes. It is typically wedge shaped and confined to the segment or segments involved. Hepatic vein occlusion, including Budd-Chiari syndrome, results in impeded venous outflow causing hepatic congestion and increased sinusoidal pressures (49). The resultant pattern of hypointensity is patchy along the drainage basin of the affected hepatic vein (Fig 16). The HB phase hypointensity represents dilated sinusoids and decreased uptake by dysfunctional hepatocytes due to increased sinusoidal pressure and/or decreased hepatofugal portal vein blood flow in the region drained by an occluded hepatic vein.

Hepatic artery thrombosis is most commonly associated with transplants, although it has many other causes. The hepatic artery is the sole supply to the biliary tree, with its occlusion placing the patient at risk for biliary tree necrosis and biloma formation as well as hepatic infarction if there is concomitant portal vein compromise (49). Despite preserved portal venous flow, regional HB phase hypointensity may be seen in patients with hepatic artery occlusion (Fig 17), probably reflecting ischemia-induced hepatocyte dysfunction. HB phase MRI is useful in these patients, allowing assessment of the integrity of the biliary tree postarterial insult, including the posttransplant liver graft (50,51).

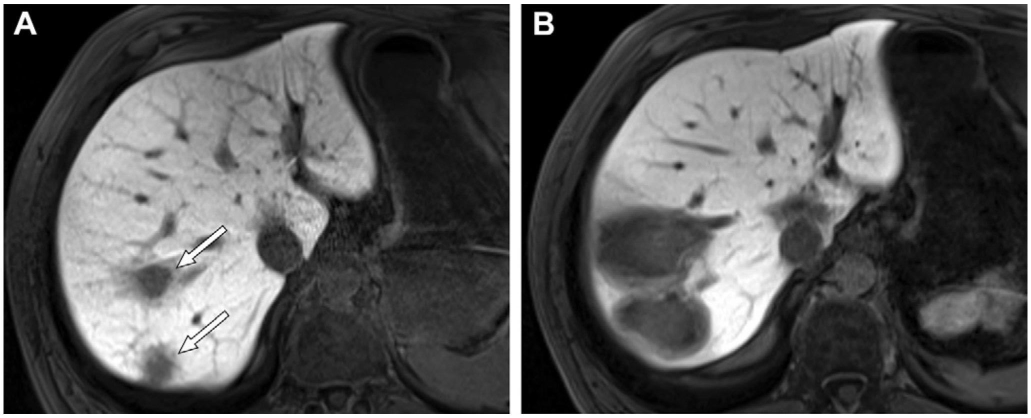

Figure 14. Colorectal carcinoma metastases to the liver in a 50-year-old man. Axial gadoxetate-enhanced HB phase MR images were acquired before **(A)** and 2 months after **(B)** microwave ablation of two colorectal metastases (arrows in **A**). The ablation zones are much larger than the treated metastases but still similar in appearance, with central hypointensity and mild peripheral hypointensity **(B)**. The apparent increase in size should not be interpreted as progression of disease. In a successfully ablated lesion, the ablation zone gradually decreases in size over months to years. During follow-up, development of an irregular outline, nodular enhancement in dynamic phases, or restricted diffusion in a nodular area should raise suspicion for recurrence.

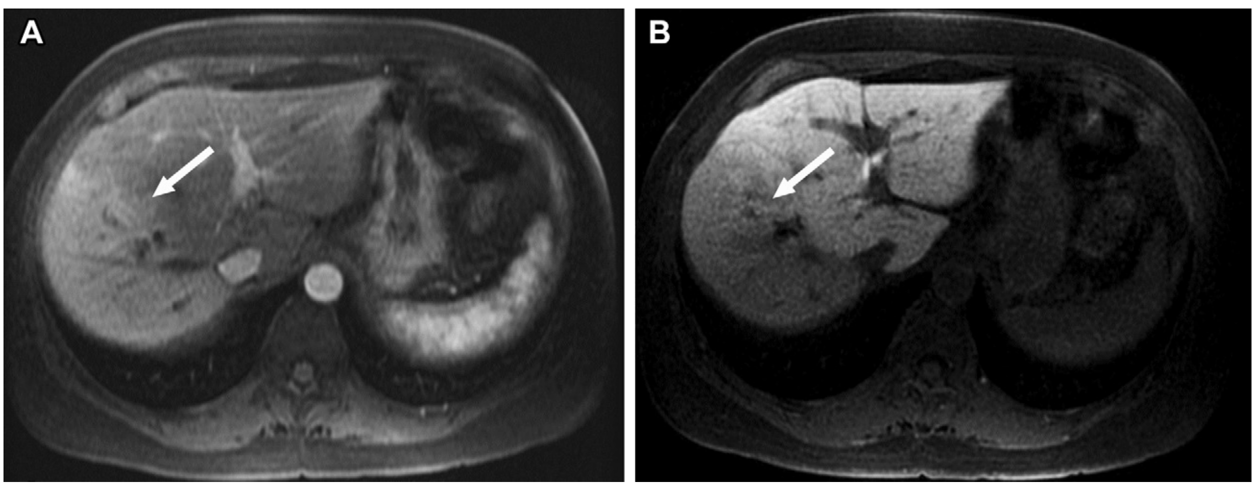

Figure 15. Colorectal metastases to the right lobe of the liver treated with right portal vein embolization in a 50-year-old man. Axial postembolization fat-suppressed T1-weighted arterial phase MR image **(A)** shows hyperenhancement of the right lobe of the liver (arrow) due to compensatory arterial flow. **(B)** Axial postembolization fat-suppressed T1-weighted HB phase MR image shows corresponding hypointensity (arrow).

Biliary Causes of Hypointensity.—Bile duct branch stenosis or occlusion results in biliary stasis in a region of the liver, and there is resultant signal hypointensity on HB phase MR images in the corresponding region of the parenchyma drained by the duct and a lack of or delayed contrast material excretion in the narrowed or occluded portion of the biliary tree (Fig 18). This may be seen regardless of the cause of the biliary stricture (52). This differential uptake may be helpful to distinguish between areas of unimpaired and regionally impaired liver parenchyma in patients with primary sclerosing cholangitis and to help guide further management (41).

Metastases and Infiltrative Diseases.—Confluent processes may involve one or more segments of the liver that appear as regions of HB phase hypointensity with vascular or biliary causes; however, these processes usually have irregular or nodular outlines. Confluent metastases (Fig S6) may also cause irregular or geographic regions of hypointensity. Amyloid deposition may vary, affecting a few segments or only one lobe, resulting in hypointensity during the HB phase of the involved region or regions.

Focal Hypointensities

Focal hypointensities may be hepatocyte-related or due to focal lesions. These show different patterns of hypointensity based on their tissue composition.

Hepatocyte-related Hypointensities.—Focal fat deposition is a benign entity that is often seen in typical locations (eg, in the fissure of the ligamentum teres or the periportal region), without substantial mass effect. Uncommonly, it can be multifocal and nodular in appearance, mimicking a

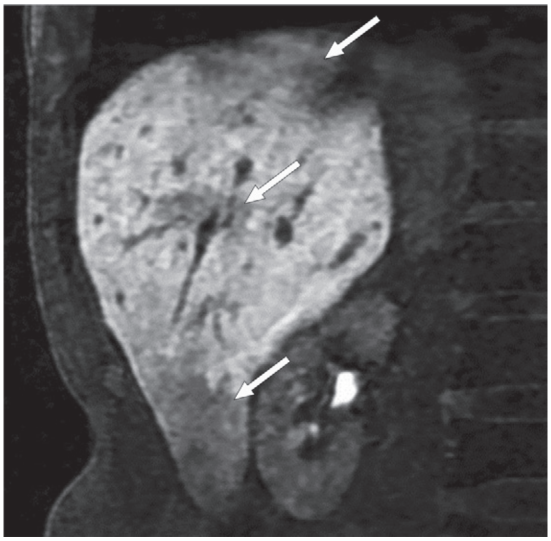

Figure 16. Colorectal carcinoma metastasis to the liver treated with radiofrequency ablation in a 55-year-old man who then developed a right hepatic vein thrombus (not shown). Coronal gadoxetate-enhanced HB phase MR image shows patchy hypointense regions (arrows) along the drainage area of the right hepatic vein, consistent with congestion.

lesion. On HB phase MR images, focal areas of parenchymal uniform hypointensity are typically seen, with relatively straight borders and without displacement of vessels or bile ducts. Furthermore, they are seen in typical locations, which helps in differentiating them from hypointense focal lesions (4) (Fig S7). Focal fat deposition is easily recognized on out-of-phase T1-weighted MR images, although all sequences must be considered to exclude an underlying fat-containing lesion (53).

Biliary Causes of Hypointensity.—Cholangitis can result in focal HB phase hypointensity (Fig 19) because of biliary stasis and inflammation (52). HB phase hypointensity usually corresponds to an abnormality in signal intensity on T2-weighted MR images. Other imaging findings may include bile duct wall thickening and enhancement. The patient may have a history of infectious signs and symptoms, primary sclerosing cholangitis, or biliary intervention. Focal abscesses, which also cause focal hypointensity, may develop (52).

Other Parenchymal Causes of Hypointensity.—Focal fibrosis will result in HB phase hypointensity. Fibrous tissue replaces hepatocytes and thus precludes hepatocyte-specific uptake of HB contrast agents.

Hypointense Focal Lesions

Benign Hepatocellular Lesions

Hepatocellular adenomas were previously thought to be exclusively hypointense during the HB phase (Fig 20), allowing their differentiation from FNHs, which were either isointense or hyperintense to the liver parenchyma, with early studies (54,55) suggesting the high specificity of this finding. However, subsequent reports (4,8,56,57) have revealed that a proportion of hepatocellular adenoma subtypes show uptake of HB contrast agents, reducing the specificity of HB phase MRI for differentiating FNHs from these hepatocellular adenoma subtypes (56). The subtypes that may demonstrate uptake include β-catenin exon-3–activated hepatocellular adenomas and inflammatory adenomas. Most FNHs (approximately 90%) are homogeneously or heterogeneously hyperintense during the HB phase, and a small proportion of FNHs (<10%) may be hypointense during the HB phase (58,59).

Malignant Hepatocellular Lesions

HCC is the most common primary liver cancer (51) and is commonly seen as hypointensity during the HB phase. However, well-differentiated HCCs (approximately 10%–15% of HCCs) may show signal hyperintensity during the HB phase. Typically, in hepatocarcinogenesis, a decrease in OATP expression levels is seen before complete neoarterialization occurs, and small HCCs may be seen as hypointense nodules, without arterial phase hyperenhancement during the HB phase. This results in the possibility of HB phase hypointensity in low- and high-grade dysplastic nodules as well as HCCs (3). Overall, HB contrast agents improve sensitivity for detection of HCCs, at the expense of specificity (4).

Careful interpretation of imaging features is required when an HB contrast agent is used for evaluation of HCCs. A targetoid appearance or hypointensity favors diagnosis of malignancy, and HB phase isointensity to the parenchyma favors benignity. There is possible decreased detection of the key features of arterial phase hyperenhancement, washout, and the presence of a capsule. Furthermore, washout can only be interpreted during the portal venous phase and not during the transitional or HB phases, while the capsule may be assessed during both the portal venous and transitional phases but not during the HB phase (3).

HB contrast agents may provide additional important diagnostic and prognostic information in patients with HCC. HB phase hypointensity is more commonly seen in poorly differentiated HCCs than it is in their well-differentiated counterparts (8). HCCs may have a hypointense rim suggestive of microvascular invasion (Fig 21), which is hypothesized to be secondary to microscopic tumor thrombi in peritumoral portal venules resulting in decreased OATPB function in hepatocytes around the tumor (60). Microvascular invasion is associated with shorter recurrence-free and overall survival times (61). Peritumoral hypointensity has also been associated with posttransplant and postresection recurrence (62,63).

Benign Nonhepatocellular Lesions

Simple cysts and hepatic hemangiomas appear hypointense on HB phase MR images. These are common and are diagnosed with conventional and contrast-enhanced MRI sequences. Hepatic abscesses also appear hypointense on HB phase images (Fig 19) and may have a faintly hyperintense rim similar to that of some metastases. Additional imaging features include central T2 signal hyperintensity, with associated parenchymal edema, peripheral enhancement, and restricted diffusion, in some cases. Hepatic abscesses vary in appearance, depending on their

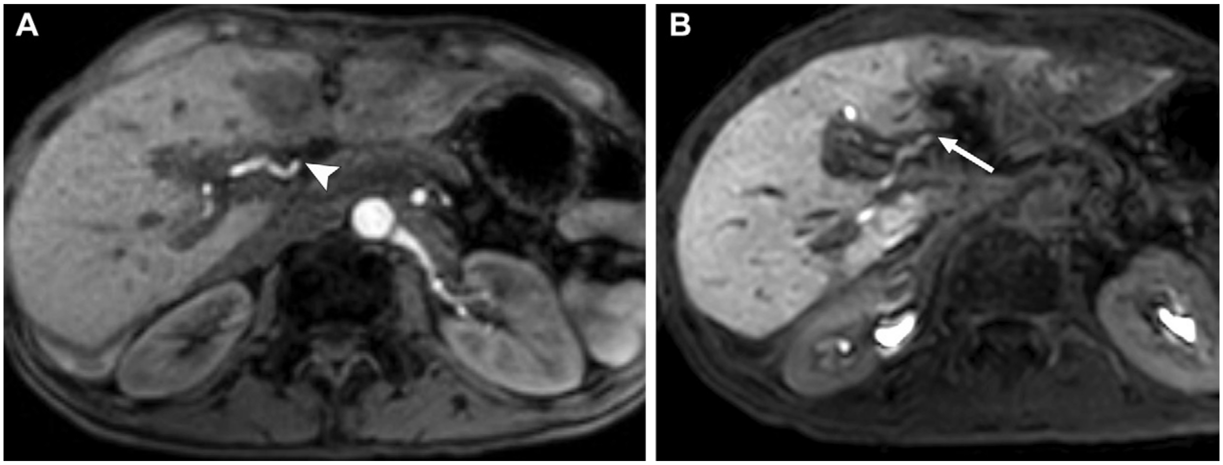

Figure 17. Acute posttransplant hepatic artery thrombosis with hepatic infarcts in a 45-year-old man with a liver transplant. **(A)** Axial dynamic arterial phase MR image shows no enhancement of the left hepatic artery (arrowhead). The portal vein was patent, and the parenchyma demonstrated normal portal venous phase enhancement (not shown). **(B)** Axial dynamic HB phase MR image shows regional hypointensity in the left lobe, but there is biliary excretion into the left hepatic duct (arrow) despite the left hepatic arterial occlusion.

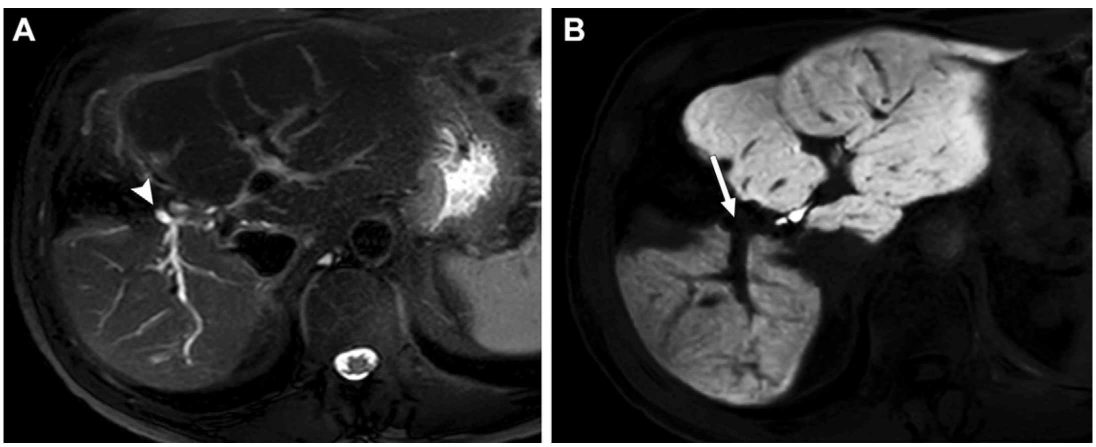

Figure 18. Right hepatic duct stricture in a 36-year-old woman who developed the stricture after undergoing partial hepatic resection for a hepatic adenoma. **(A)** Axial T2-weighted MR image shows a proximal right hepatic duct stricture (arrowhead) with upstream biliary and associated increased T2 signal intensity throughout the residual right hepatic lobe. **(B)** Axial T2-weighted HB phase MR image shows a corresponding regional hypointensity as well as nonopacification of the right biliary tree (arrow) in comparison with the appearance of the left biliary tree.

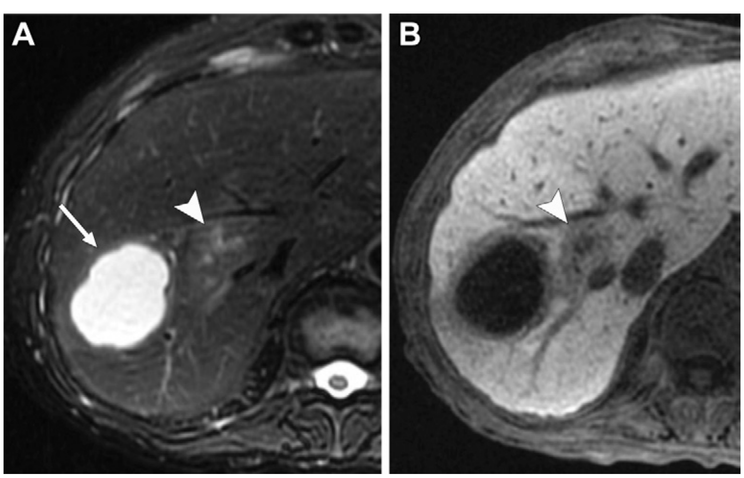

Figure 19. Hepatic abscess and cholangitis in a 68-year-old woman with a history of metastatic lung carcinoma who underwent bilateral adrenalectomy with adrenal insufficiency and was currently undergoing steroid treatment. The patient also had a history of a perforated duodenal ulcer, a cholecystoduodenal fistula, recurrent cholangitis, and liver abscesses and presented with a fever. **(A)** Axial T2-weighted MR image shows a hyperintense abscess cavity (arrow) in the right lobe of the liver adjacent to faint signal hyperintensity secondary to cholangitis (arrowhead). **(B)** Axial T2-weighted HB phase MR image shows the abscess cavity as hypointense, with a mildly hyperintense rim, which has signal intensity less than that of the adjacent liver parenchyma. The adjacent cholangitis is seen as hypointensity (arrowhead) and corresponds to the hyperintensity seen in **A**. Focal biliary ductal dilatation of the right hepatic duct was present (not shown). The abscess was treated with antibiotics and subsequent percutaneous drainage.

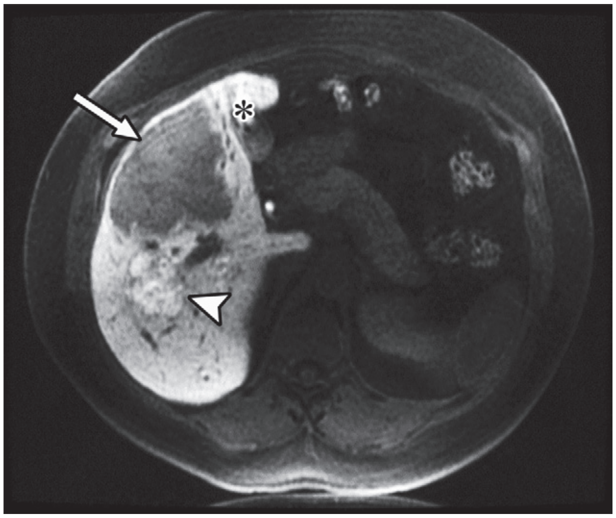

Figure 20. Hepatic adenoma versus FNH in a 32-year-old woman with obesity and a history of oral contraceptive use for more than 15 years, with multiple hepatic lesions. Axial gadoxetate-enhanced HB phase MR image shows two large lesions in the right lobe: a large subcapsular lesion (arrow) and a lobulated hyperintense lesion (arrowhead), as well as contrast material in the gallbladder (*). The patient underwent right hepatectomy because biopsy of the larger hypointense lesion did not rule out the possibility of well-differentiated HCC and because of the possibility of rupture of the large subcapsular lesion. Histologic results revealed that the larger hypointense lesion was a benign β-catenin–negative inflammatory adenoma and the smaller hyperintense lesion was an FNH.

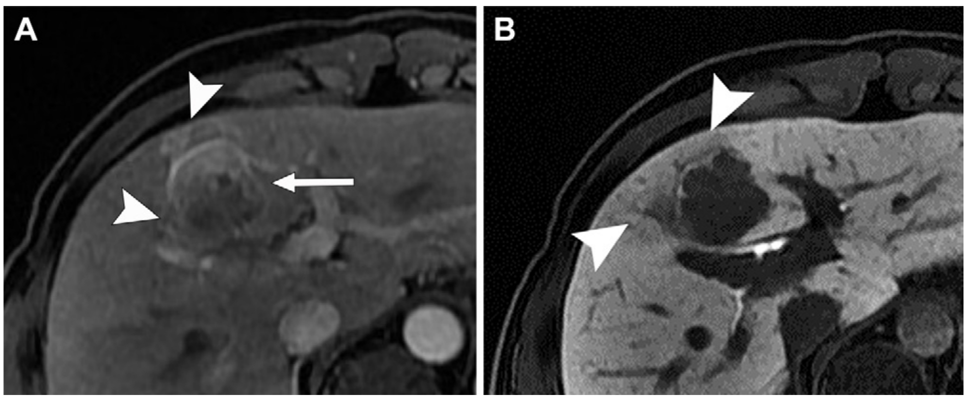

Figure 21. HCC in a 46-year-old man with chronic hepatitis B. **(A)** Axial gadobenate-enhanced fat-saturated T1-weighted arterial phase MR image shows nonrim arterial phase hyperenhancement (arrow), with irregular peritumoral enhancement (arrowheads). **(B)** Axial gadobenate-enhanced fat-saturated T1-weighted HB phase MR image shows that the mass is predominantly hypointense but also has faint irregular peritumoral hypointensity (arrowheads), which is suggestive of microvascular invasion.

stage and/or treatment. Other imaging sequences and a clinical history of infectious symptoms may be helpful in diagnosis.

Malignant Nonhepatocellular Lesions

Cholangiocarcinoma is the second most common primary liver cancer. Although they are a heterogeneous group of tumors, cholangiocarcinomas have abundant fibrous stroma and numerous capillaries at pathologic examination (64). Cholangiocarcinomas are predominantly hypointense during the HB phase, but the central scirrhous part can retain HB contrast agent in the extracellular space associated with the desmoplastic reaction or necrosis (65,66). This gives rise to the "target" sign, with mild central hyperintensity that is lower than that of the background parenchyma and peripheral hypointensity (Fig 22). The sign is not specific for cholangiocarcinoma but is also seen in other tumors with a desmoplastic reaction such as colorectal metastases (Fig 23A) (67,68).

Despite the possibility of HB contrast agent uptake into metastases, the HB phase retains substantial value in identifying metastases as hypointense lesions (69). Metastases commonly manifest as diffusely hypointense on HB phase MR images, without uptake from baseline, with some variation. Mild retention may be seen centrally in metastases with fibrotic stroma or possibly aberrant OATP expression in tumor cells that is incompletely understood at this time (70). Wedge-shaped or irregularly shaped areas of hypointensity around a lesion may result from arterioportal shunting and resultant decreased portal venous flow to the area (70,71). Some metastases including renal cell carcinoma, neuroendocrine tumors, gastrointestinal stromal tumors, and colorectal adenocarcinomas may show a homogeneous thin hyperintense rim (Fig 23B) that is postulated to be secondary to compression of veins, leading to hypertrophy of the perivenular hepatocytes (72).

Implications

The use of HB contrast agents and their uptake as a biomarker of liver function have been correlated with better outcomes for the patient and more accurate prognostication (5,50,73). Accurate detection of hypointensity can be useful information for planning future treatment in patients with liver tumors.

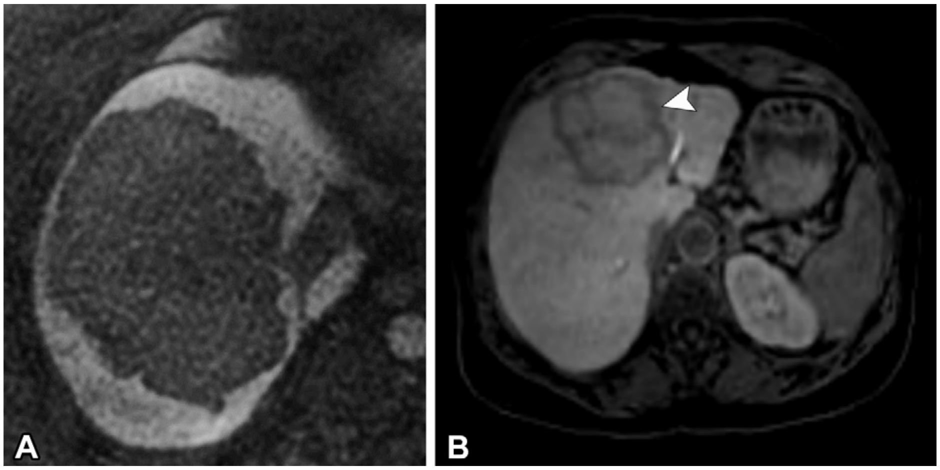

Figure 22. Intrahepatic cholangiocarcinoma in two patients. **(A)** Axial HB phase fat-suppressed T1-weighted MR image shows a large intrahepatic cholangiocarcinoma as completely hypointense. **(B)** Axial HB phase fat-suppressed T1-weighted MR image in a different patient shows an intrahepatic cholangiocarcinoma with peripheral rim–like hypointensity (arrowhead) and central cloudlike hyperintensity due to retention of HB contrast material in the extracellular fluid, termed the "target" sign.

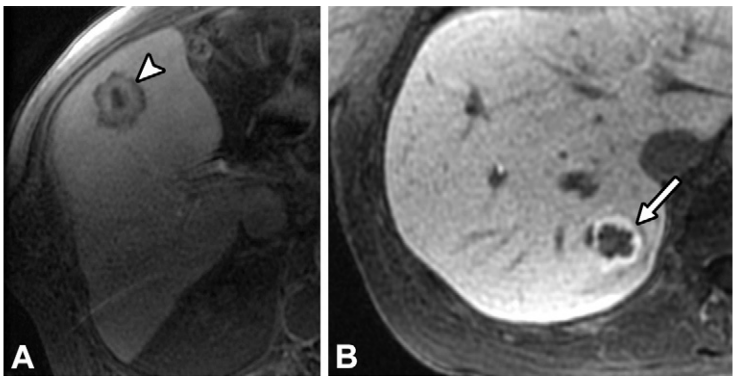

Figure 23. Variable appearances of hepatic metastases in two patients. **(A)** Axial HB phase fat-suppressed T1-weighted MR image in a 66-year-old man with colorectal adenocarcinoma metastases shows the metastasis (arrowhead) with alternating signal hypointensity and hyperintensity in a layered appearance due to contrast material retention in the core except for a central necrotic region, referred to as the target sign. **(B)** Axial HB phase fat-suppressed T1-weighted MR image in a 72-year-old man with a history of having undergone nephrectomy for clear cell renal cell carcinoma shows a metastasis with a hypointense core and a hyperintense rim (arrow). Multiple liver metastases were present. The lesion was biopsied and confirmed to be a metastatic clear cell renal carcinoma. The two other liver metastases had similar appearances of a hypointense core and hyperintense rim.

Pitfalls

Although a substantial amount of information may be determined from HB phase MR images, it is possible to miss or misdiagnose findings when HB phase images are interpreted in isolation. Determining the quality of the examination and ensuring that there are no technical causes of hypointensity is important. In patients with diffuse hypointensity on HB phase MR images due to steatosis, iron overload, or chronic liver disease, sensitivity for detection of a lesion may be decreased (Fig 24). Lesion detection is also decreased if there is regional hypointensity, which may obscure an underlying abnormality (Fig 25). All MRI sequences must be considered in assessment of a lesion to make an accurate diagnosis and measurement. If HB phase hypointensity is the primary consideration in diagnosing a lesion, key features leading to appropriate diagnosis may be missed. For example, although FNH is commonly hyperintense during the HB phase there are rare instances of FNH showing hypointensity. Misdiagnosis can be avoided with assessment for key features on images from additional sequences (Fig S8). Similarly, the hyperintense or hypointense rim around hepatic observations can lead to overmeasuring or undermeasuring and incorrect assessment of disease status. Overall, most pitfalls of HB phase imaging may be avoided by correlating findings of all sequences for a complete assessment. A stepwise approach to evaluating HB phase hypointensity is summarized in the flowchart (Fig 26).

Conclusion

Hypointensity during the HB phase has many causes and numerous pathophysiologic origins. The pattern of hypointensity—diffuse, regional, or focal—allows narrowing of the differential diagnosis. Integrating clinical information and all MRI sequences is paramount to diagnostic accuracy.

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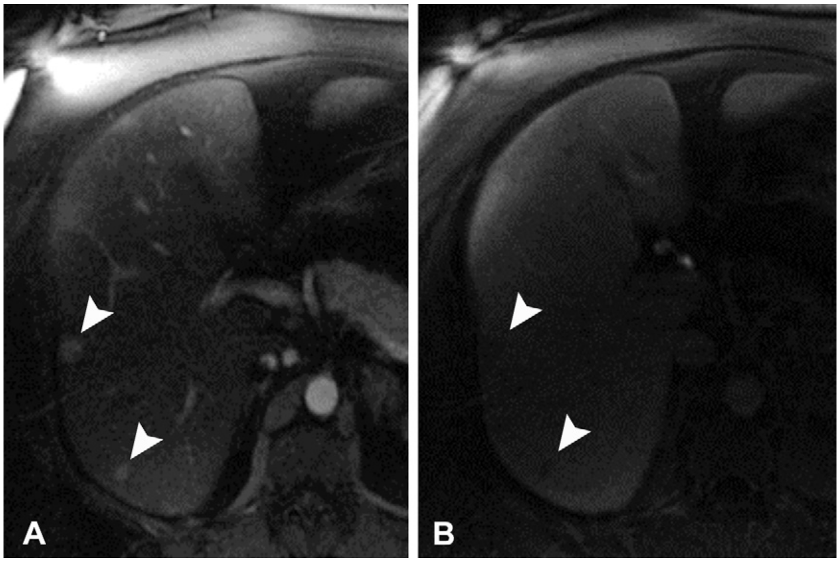

Figure 24. Hepatic steatosis due to metabolic dysfunction in a 32-year-old woman with obesity, associated steatotic liver disease, and a hepatic proton density fat fraction of 45%. **(A)** Axial arterial phase fat-suppressed T1-weighted MR image shows two enhancing lesions in the liver periphery (arrowheads). **(B)** Axial HB phase fat-suppressed T1-weighted MR image shows diffuse hypointensity of the liver parenchyma, obscuring the underlying observations with only the anterior lesion barely seen (arrowheads). The patient underwent gastric sleeve surgery, lost more than 120 lb (53.43 kg), and the proton density fat fraction improved to 25%. The lesions were stable at follow-up after 4 years and were presumed to be hepatic adenomas.

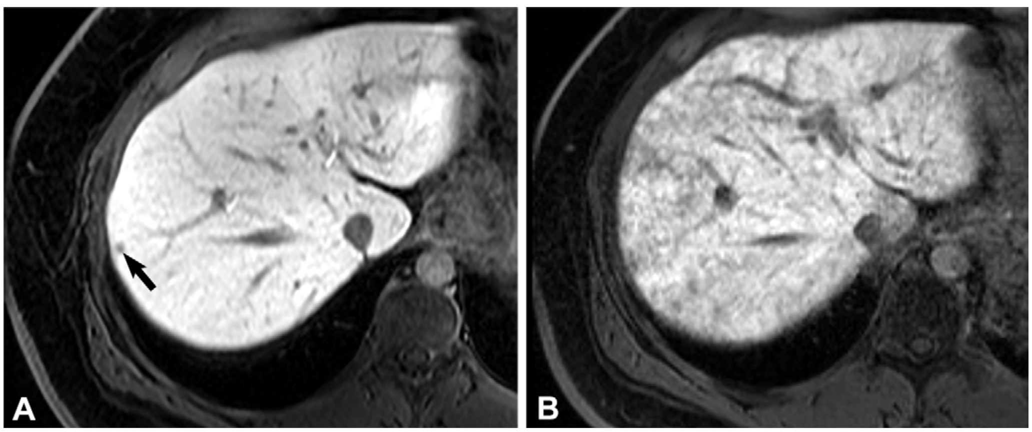

Figure 25. SOS in a 52-year-old man with metastatic cholangiocarcinoma who was currently undergoing chemotherapy. Axial fat-suppressed T1-weighted MR images acquired during the arterial phase **(A)** and HB phase **(B)** show that the hepatic metastasis is clearly delineated (arrow in **A**), but the conspicuity is decreased on the HB phase image **(B)** due to the peripheral and patchy reticular hypointensities from SOS.

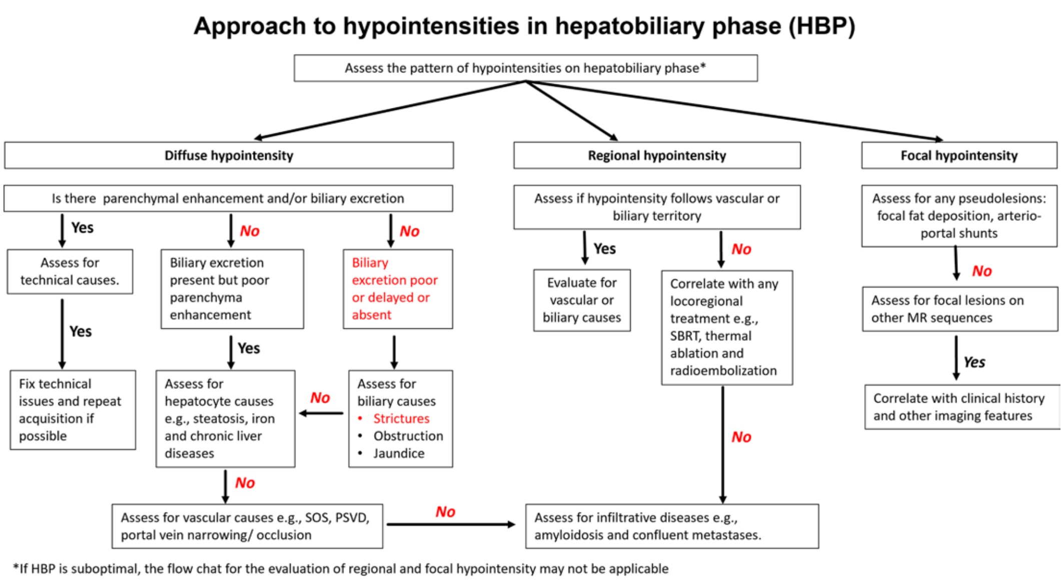

Figure 26. Flowchart shows the approach to evaluating hypointensities during the HB phase *(HBP)*.

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